Assignment 7: Epistasis & Gene Linkage

Task 2

Triple Epistasis (Task 2)

Estimation Formulas

Baseline 1

In the **first-order approach**, the null model $\mathcal{H}_0^1$ assumes independence and additive decomposability of individual fitness effects:

$$\mathcal{H}_0^1: \quad \Delta F_{ijk} = \Delta F_i + \Delta F_j + \Delta F_k$$

The triplet epistasis measures the total interaction deviation from additivity:

$$\varepsilon_{ijk}^{(1)} := \underbrace{\left[F(g \setminus \{i,j,k\}) - F(g)\right]}_{\Delta F_{ijk}} - (\Delta F_i + \Delta F_j + \Delta F_k) \equiv F(g \setminus \{i,j,k\}) - F(g \setminus \{i\}) - F(g \setminus \{j\}) - F(g \setminus \{k\}) + 2F(g)$$

Baseline 2

In the **second-order approach**, the null model $\mathcal{H}_0^2$ accounts for both single-gene knockouts ΔF_i and all pairwise epistatic interactions between the three genes ε_{ij}:

$$\mathcal{H}_0^2: \quad \Delta F_{ijk} = \Delta F_i + \Delta F_j + \Delta F_k + \underbrace{\varepsilon_{ij} + \varepsilon_{ik} + \varepsilon_{jk}}_{\text{2nd-order effects}} \equiv \Delta F_{ij} + \Delta F_{ik} + \Delta F_{jk} - (\Delta F_i + \Delta F_j + \Delta F_k)$$

This first expression is analogous to the exclusion-inclusion principle, which generalizes the definition of n-th order epistasis $\varepsilon^{(n-1)}$.

Hence epistasis is calculated as:

$$\varepsilon_{i,j,k}^{(2)} := \underbrace{\Delta F_{ijk} - \left(\Delta F_i + \Delta F_j + \Delta F_k\right)}_{\downarrow} - \left(\varepsilon_{ij} + \varepsilon_{ik} + \varepsilon_{jk}\right)$$

$$\varepsilon_{i,j,k}^{(2)} \equiv \varepsilon_{i,j,k}^{(1)} - \left(\varepsilon_{ij} + \varepsilon_{ik} + \varepsilon_{jk}\right)$$

- $\varepsilon_{i,j,k}^{(2)} = 0 \implies$ pairwise decomposability: the interaction of genes is predictable from pairwise interactions, no higher-order effects.
- $|\varepsilon_{i,j,k}^{(2)}| > 0 \implies$ gene k modulates interaction between i and j.

Note: *In case of Baseline 1, $\varepsilon_{i,j,k}^{(1)} = 0$ would mean either that the interaction is strictly additive $(\Delta F_{ijk} = \Delta F_i + \Delta F_j + \Delta F_k)$ or that unaccounted higher-order effects cancel out existing pairwise ones $(\varepsilon_{i,j,k}^{(2)} = -(\varepsilon_{ij} + \varepsilon_{ik} + \varepsilon_{jk}))$. Hence, the definition of $\varepsilon^{(1)}$ yields less distinct information than $\varepsilon^{(2)}$. In terms of **new** information, Baseline 2 explicitly factors out known pairwise effects, while Baseline 1 leaves them entangled with possible triplet effects.*

In the calculated 3D tensors $(L \times L \times L)$ and Figure 5:

- The main diagonal ($i = j = k$) stores the single-gene knockout effect $\Delta F_i (\equiv \Delta F_{iii})$.
- Diagonal planes with 2 coincide indices (e.g. $(i = j, k)$) store pairwise epistases $\varepsilon_{i,j}$.
- The off-diagonal entries ($i \neq j \neq k \neq i$) store the triplet epistasis ($\varepsilon_{ijk}^{(\cdot)}$). For a given knocked out gene k, a cross-section parallel to one of the coordinate planes - $\pi_{ij}^k = \{(i,j) | \text{fix } k\}$ - stores the pairwise epistases ε_{ij}^k after gene k is knocked out.

3D Epistasis Heatmaps

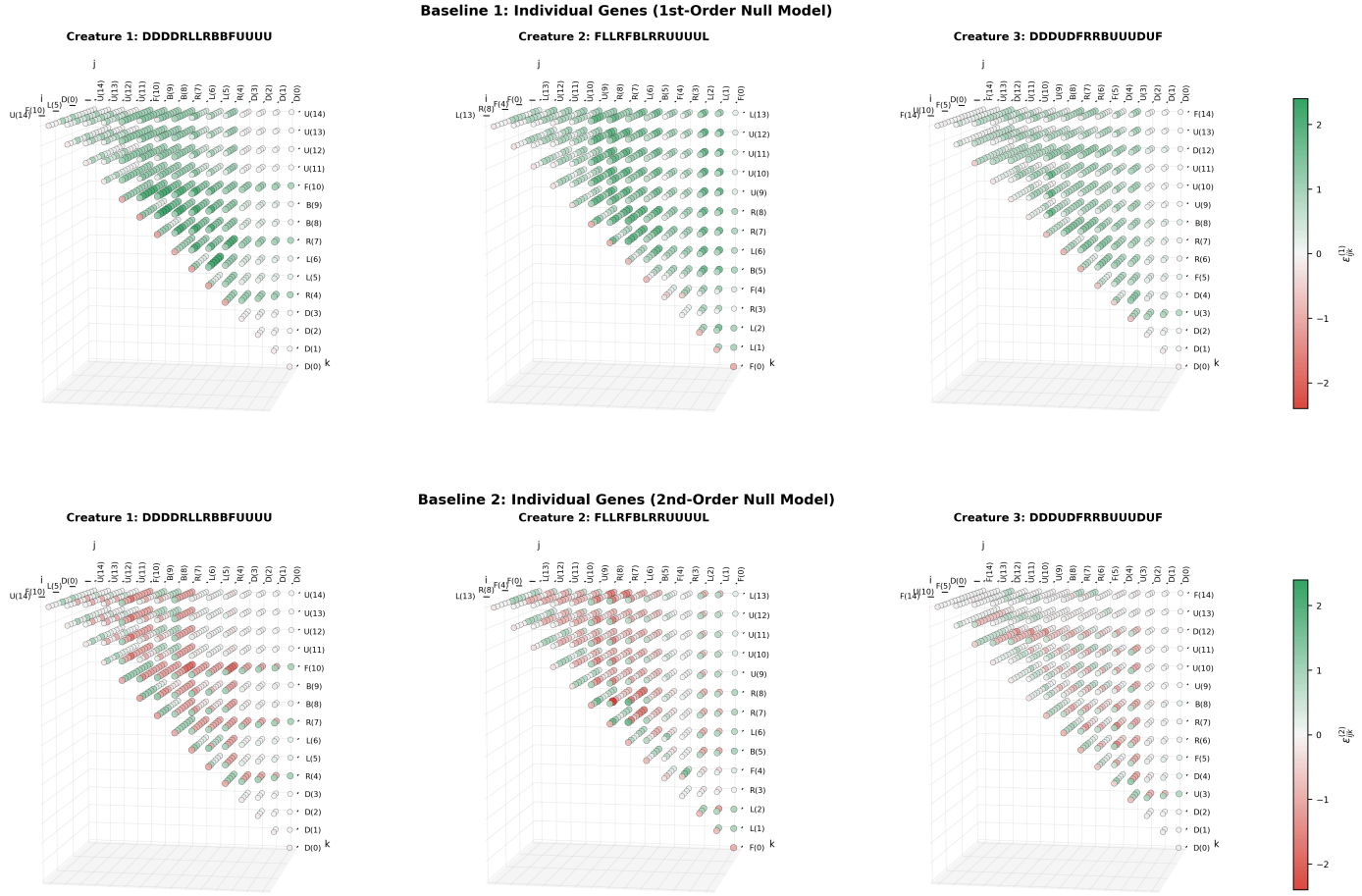


Figure 5: *3D epistasis heatmaps (balls arranged in a 3D grid) for $k = 3$ dissimilar creatures, depicting $\varepsilon^{(\cdot)}_{ijk} \mid \forall i \leq j \leq k$.* **Top row:** Baseline 1 (relative to individual genes_ $\varepsilon^{(1)}_{ijk}$). ***Bottom row:*** *Baseline 2 (relative to pairwise interactions $\varepsilon^{(2)}_{ijk}$).* ***Colormap:*** *Green - positive synergy / buffering; Red - negative / synthetic breakdown. The scale of the colormap differs from the one used in 2D-epistatic visualisations. A semiplane on the visible side of the pyramids coincides with the upper triangle of epistasis matrices (Figure 4).*

Epistasis Matrix Sums Comparison

To better understand the relative contribution of pairwise and triplet epistasis, the following two metrics can be calculated:

- **Directedness** - ratio of the sum of all triplet epistasis values to the sum of the absolute values of all triplet epistasis values. Values close to ± 1 indicate that the contribution of triplet epistasis is predominantly synergistic/buffering or antagonistic/disruptive, respectively:

$$\operatorname{dir} \varepsilon^{(\cdot)} := \frac{\sum \varepsilon^{(\cdot)}}{\sum |\varepsilon^{(\cdot)}|} \in [-1; 1]$$

- **Mean absolute triplet epistasis** (this value should be considered with respect to $\max \Delta F$ which is ommited for simplification):

$$|\bar{\varepsilon}^{(\cdot)}| := \frac{1}{\binom{L}{3}} \sum |\varepsilon^{(\cdot)}|$$

Creature	Genotype	L	Triplet Count $\binom{L}{3}$	$\sum \varepsilon^{(1)}$	Baseline $\sum \lvert\varepsilon^{(1)}\rvert$	1 $\operatorname{dir}\varepsilon^{(1)}$	$\lvert\bar{\varepsilon}^{(1)}\rvert$	$\sum \varepsilon^{(2)}$	Baseline $\sum \lvert\varepsilon^{(2)}\rvert$	2 $\operatorname{dir}\varepsilon^{(2)}$	$\lvert\bar{\varepsilon}^{(2)}\rvert$
1	`/*9*/DDDDRLLRBBFUUUU`	15	455	**+375.02**	**377.74**	+0.993	0.830	**-171.78**	224.35	-0.766	0.493
2	`/*9*/FLLRFBLRRUUUUL`	14	364	+344.29	344.55	**+0.999**	**0.947**	-149.80	183.42	**-0.817**	0.504
3	`/*9*/DDDUDFRRBUUUDUF`	15	455	+274.31	276.66	+0.992	0.608	-62.25	**142.43**	-0.437	**0.313**

Interpretation of 3D Triplet Epistasis

1. **Baseline 1 vs Baseline 2**:
   - **Baseline 2** turned out to be distinctly more useful for discovering new higher-order information regarding gene triplets.
   - Under **Baseline 1**, the null model $\mathcal{H}_0^1$ assumes pure additivity from single-gene knockouts. When evaluated against independent single-gene effects, all creatures exhibit overwhelming positive epistasis (nearly all green balls, $\mathrm{dir}\varepsilon^{(1)} \approx +1$).
   - This occurs because removing any single *base* gene (e.g. anchor `R`^{4}, symmetry genes `LL`$^{5\text{-}6}$, or stabilizer `F`^{10} in Creature 1) causes the structure to topple over ($\Delta F \approx -1.0$). *Summing three such single knockout penalties predicts an impossible fitness penalty* of -3.0, whereas the simultaneous triple knockout cannot reduce fitness below ground level ($F = 0$). Thus, the actual **fitness is buffered against cumulative linear collapse** ($\Delta F_{ijk} \gg \Delta F_i + \Delta F_j + \Delta F_k$).
   - This suggests that even in such simple (f_9) creatures the hypothesis about triple linearity $\mathcal{H}_0^1$ (see Baseline 1) does not hold and **Baseline 1 model is not adequate and yields little information** (excluded pairwise effects, almost constant directedness $\mathrm{dir}\ \varepsilon^{(1)} \approx 1$) *in the current setup*.
2. **Baseline 2 Mechanics** in Creatures 1 and 2:
   - Once all pairwise interactions are accounted for, the total epistasis flips to negative (prominent red clusters, $\mathrm{dir}\ \varepsilon^{(2)} \ll 0$).
   - As established in Task 1, ***base* genes have strong pairwise collaborations** with each other and intermediate collaboration with the *spine* encoding genes. When one of *base* genes b is knocked out, in most cases creature topples over and the further effects can be investigated across several regimes:
     - **+ 2 spinal genes** (knockout triplet b, s_1, s_2): no additional damage beyond initial toppling (main terms in the epistasis formula are $\Delta F_{b,s_1,s_2} - \Delta F_b \approx 0 \implies$ white/neutral bars on 3D map);
     - **+ base + spinal gene** (triplet b, b', s): dropping an additional *spinal* gene s produces diminishing marginal damage compared to what the pairwise sum predicts for (b, b'), manifesting as negative third-order interaction (due to 2 individual base effects $\Delta F_{b,b',s} - (\Delta F_{b'} + \Delta F_b) \ll 0$).
     - **Buffering/Rescue**:
       - In specific cases, knocking out two base genes buffers against dropping a third gene, preserving stability. For instance, in creature 1 knocking out any of genes `0-3` once genes `5-6` are removed, preserves stability of the creature - green row on a 3D map.
       - An analogous type of buffering happens in the same creature when *base* gene `8/9` is knocked out and then 2 "parallel" spinal genes (`0-3` and `11-14`) must be dropped out to preserve stability of the creature.
     - **+ 2 base genes**: Similar to the effect of knocking out 2 bases, this may yield the strong negative $\varepsilon^{(2)}$ value in case of strong pairwise dependency: knocking genes `6-8` in Creature 2 yields the formula with multiple dominating negative terms ($-\varepsilon_{..}$) in the epistasis formula.
3. **Informational value of signs in Baseline 2**:
   - **Positive epistasis** (*greenish clusters* on Figure 5) - provides genuine **new higher-order structural information**. These points reveal true 3-way buffering / rescue effects where simultaneous knockouts preserve posture or prevent fatal toppling that occurs in single or pairwise mutations (e.g., knocking out genes `3-4` alongside `1` or `2` in Creature 3 retains stability).
   - **Negative epistasis** (*reddish clusters*), on the other hand, largely reflects **damage saturation at the fitness floor** ($F \geq 0$) rather than novel triplet mechanics. When one of *base* genes is knocked out, in most cases each individual and pairwise mutation already causes total collapse ($\Delta F \approx -1.0$). Because the pairwise model registers large positive buffering or intermediate interaction, Baseline 2 subtracts these positive pairwise terms from a saturated ΔF_{ijk}, mechanically yielding negative values ($\varepsilon^{(2)} \ll 0$). Hence, **red values mostly restate the pairwise fragility of the base rather than discover a new physical interaction**. *See reddish triangles ($k = 7, 10$ in Creature 1).*
   - **Neutral values** (*whitish clusters*) signify exact pairwise decomposability (e.g., one *base* knockout causing collapse while two neutral *spine* knockouts produce no further effect).
4. **Creature 3**: Modularity and low triplet epistasis:
   - Features the lowest magnitude of 3-way epistasis ($|\sum \varepsilon^{(2)}| = 62.25$ vs 171.78 in Creature 1).
   - Its alternating loops (`DD[DUD]FRRBUU[UDU]F`) distribute structural stabilization across several self-canceling sub-segments (`UD` , `DU`), making its higher-order genetic interactions more modular, which along with simple topology of the creature (snake-like), suggests that genome of this creature features the least amount of triple epistatic effects.